

- Q.27 Describe the components of a feed system.
- Q.28 Explain balancing of runners in a feed system.
- Q.29 Describe the function of ejector plate assembly.
- Q.30 Explain the types of ejector pins used in moulds.
- Q.31 What are the advantages of cooling inserts in moulds.
- Q.32 Explain water connections in cooling systems.
- Q.33 Write a short note on side core and side cavity moulds.
- Q.34 What is a split mould? Explain briefly.
- Q.35 Describe briefly threaded component moulds.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain with neat diagrams the different types of moulds used in industry.
- Q.37 Describe the construction and working of an ejector plate assembly with a diagram.
- Q.38 Explain the cooling system in detail with types and working.

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5th Sem / Plastic, Chem Engg. (Spl. Polymer Engg.) Subject:- Design of Dies and moulds - I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which plate is used to support the die components?
- a) Bolster plate b) Cavity plate
c) Core plate d) None
- Q.2 Which type of mould has three plates?
- a) Split mould b) Runnerless mould
c) Three plate mould d) Insert mould
- Q.3 Which part guides the movement of the core and cavity?
- a) Guide bush b) Ejector plate
c) Runner d) Sprue
- Q.4 Which system balances the flow of molten plastic in a mould?
- a) Feed system b) Cooling system
c) Ejection system d) Venting system

- Q.5 Which part removes the moulded component from the cavity?
- a) Runner b) Ejector pin
c) Sprue d) Cooling channel
- Q.6 Which type of gate is commonly used for thin-walled products?
- a) Pin gate b) Fan gate
c) Edge gate d) Ring gate
- Q.7 Which cooling medium is generally used in moulds?
- a) Water b) Steam
c) Air d) Oil
- Q.8 What is the main function of the runner in a feed system?
- a) Provide cooling b) Guide molten plastic
c) Eject product d) Vent gases
- Q.9 Which mould type is used for threaded components?
- a) Side core mould b) Two plate mould
c) Insert mould d) None
- Q.10 Which component ensures proper alignment of mould halves?
- a) Sprue bush b) Guide pin
c) Cooling channel d) Runner

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define mould.
- Q.12 Name any two types of impressions in moulds.
- Q.13 What is the function of a core?
- Q.14 Write two types of parting surfaces.
- Q.15 What is the purpose of a sprue?
- Q.16 Write any two types of gates.
- Q.17 What is the function of the cooling system in a mould?
- Q.18 Name any two types of ejectors.
- Q.19 Write one use of runnerless moulds.
- Q.20 What is meant by multi-daylight moulds?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List any five types of moulds and their uses.
- Q.22 Explain briefly the types of cavities used in moulds.
- Q.23 Write short notes on guide pillars and guide bushes.
- Q.24 What are the functions of a bolster plate?
- Q.25 Explain the selection criteria for parting surfaces.
- Q.26 List any five types of gates and their applications.